

Solutions to the guided practice. Section numbers and lettered parts match the student handout.

1. Random variables, realizations, and averages

- a. X is a random variable; x denotes one possible realized value. Applying g gives the random variable $g(X)$, whereas $\mathbb{E}[g(X)]$ is a fixed average determined by the distribution. When $x = 3$, the realized value is $g(3) = 9$.
- b. The weighted sums are

$$\mathbb{E}[X] = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{3} + 3 \cdot \frac{1}{6} = \frac{5}{6}, \quad \mathbb{E}[X^2] = 0^2 \cdot \frac{1}{2} + 1^2 \cdot \frac{1}{3} + 3^2 \cdot \frac{1}{6} = \frac{11}{6}.$$

Thus $\mathbb{E}[X^2] = 11/6$, while $(\mathbb{E}[X])^2 = 25/36$; they are different.

2. CDFs, densities, and expected payments

- a. The density is $f_X(x) = 2x$ on $(0, 1)$ and zero outside $[0, 1]$. Values of the density at individual endpoints do not affect any integral. We have

$$\Pr[X > 1/2] = 1 - F_X(1/2) = 1 - \frac{1}{4} = \frac{3}{4}, \quad \Pr[X = 1/2] = 0.$$

- b. The realized payment is $g_r(X) = r\mathbf{1}\{X \geq r\}$. Thus

$$\mathbb{E}[g_r(X)] = r \Pr[X \geq r] = r(1 - r^2).$$

The integral check gives $\int_r^1 r(2x) dx = r[x^2]_r^1 = r(1 - r^2)$. The weights come from the value's density, and the output being averaged is the payment.

3. Conditioning and the tower property

- a. The row totals are $1/2, 1/2$, and the column totals are $5/8, 3/8$. Hence

$$\Pr[Y = 1 \mid X = 0] = \frac{1/8}{1/2} = \frac{1}{4}, \quad \Pr[Y = 1 \mid X = 1] = \frac{1/4}{1/2} = \frac{1}{2}.$$

- b. Since $Y \in \{0, 1\}$, $m(0) = 1/4$ and $m(1) = 1/2$. Thus $m(X)$ takes values $1/4$ and $1/2$, each with probability $1/2$. The outer average is

$$\mathbb{E}[m(X)] = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{8}.$$

This matches $\mathbb{E}[Y] = 0 \cdot (5/8) + 1 \cdot (3/8) = 3/8$.

- c. They are not independent: $\Pr[X = 1, Y = 1] = 1/4$, but $\Pr[X = 1] \Pr[Y = 1] = (1/2)(3/8) = 3/16$. One failed equality suffices to disprove independence.

4. Fix a value, then average; reverse the bounds

- a. Independence leaves Y uniform after fixing $X = x$, so

$$m(x) = \int_0^x y \, dy = \frac{x^2}{2}, \quad \mathbb{E}[W] = \int_0^1 \frac{x^2}{2} \, dx = \frac{1}{6}.$$

The inner average is over Y with x fixed; the outer average uses the density 1 of X on $[0, 1]$.

- b. At a fixed $y \in [0, 1]$, the allowed values are $y \leq x \leq 1$. Therefore

$$\mathbb{E}[W] = \int_{y=0}^1 \int_{x=y}^1 y \, dx \, dy = \int_0^1 y(1-y) \, dy = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Both orders integrate the same function over the same triangular region.

5. Tails, surplus, and independent events

- a. On $[0, 1]$, the tail probability is $\Pr[X_2 > t] = 1 - t$. Hence

$$\mathbb{E}[(X_2 - T)^+] = \int_T^1 (1 - t) \, dt = \frac{1}{2} - T + \frac{T^2}{2} = \frac{(1 - T)^2}{2}.$$

Also $\Pr[A] = T$. The indicator $\mathbf{1}\{A\}$, determined by X_1 , is independent of $(X_2 - T)^+$, determined by X_2 . Thus

$$\mathbb{E}[\mathbf{1}\{A\}(X_2 - T)^+] = \frac{T(1 - T)^2}{2}.$$

- b. On C , the indicator is one. Outside C , the surplus is already zero. Therefore the product equals the surplus on every outcome, and its expectation is $(1 - T)^2/2$. But

$$\Pr[C]\mathbb{E}[(X_2 - T)^+] = (1 - T)\frac{(1 - T)^2}{2} = \frac{(1 - T)^3}{2},$$

which is smaller for $0 < T < 1$. The event and the surplus are dependent.

6. Order statistics: the largest and second-largest draws

- a. All three draws must be at most t , so independence gives $F_M(t) = t^3$ for $0 \leq t \leq 1$. Hence

$$\mathbb{E}[M] = \int_0^1 (1 - t^3) dt = 1 - \frac{1}{4} = \frac{3}{4}.$$

The CDF is zero for $t < 0$ and one for $t > 1$.

- b. There are three choices for the unique draw above t . The three cases are disjoint, so

$$F_S(t) = t^3 + 3(1 - t)t^2 = 3t^2 - 2t^3, \quad f_S(t) = 6t - 6t^2 = 6t(1 - t)$$

on $(0, 1)$. Therefore

$$\mathbb{E}[S] = \int_0^1 t 6t(1 - t) dt = 6 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{1}{2}.$$

- c. After fixing the realized values, the only remaining randomness is the deleted index. Each index still has probability $1/3$ because deletion is independent of the values. Deleting the index of the largest draw leaves maximum s ; deleting either of the other two indices leaves maximum m . Thus

$$\mathbb{E}[R \mid X_1 = x_1, X_2 = x_2, X_3 = x_3] = \frac{2}{3}m + \frac{1}{3}s.$$

Averaging over the draws and using linearity gives

$$\mathbb{E}[R] = \frac{2}{3}\mathbb{E}[M] + \frac{1}{3}\mathbb{E}[S] = \frac{2}{3} \cdot \frac{3}{4} + \frac{1}{3} \cdot \frac{1}{2} = \frac{2}{3}.$$

This agrees with the expected maximum of two independent uniform draws: $\int_0^1 (1 - t^2) dt = 2/3$. Random deletion is independent of the values, so the remaining draws still have this distribution.